

Mathematics

A Cultural Heritage

Lecture 8 (part 1)

Theorems

Course Topics

*Funny Side of
Mathematics*

Mathematics,
literatures, movies,
and more

**Mathematics
and the Arts**

**Mathematics
and Education**

Two mathematics
cultures and a few
mathematical lives

Numbers

Numbers
at work

Our Obsessions
with Numbers

**Mathematics
and Society**

Mathematics and Truth
(Philosophical Aspects
of Mathematics)

**THEOREMS (PROBLEMS)
AND PROOFS**

Lecture Outline

1. What are Theorems?
2. What are Proofs? Why do we need them?
3. Some Famous Theorems

Truth and Theorems

1. Truth is what is really true.
2. Theorems are the things that we can prove are true.
3. Gödel proved that these may be different.

The First Incompleteness Theorem

In any axiomatic mathematical system which is

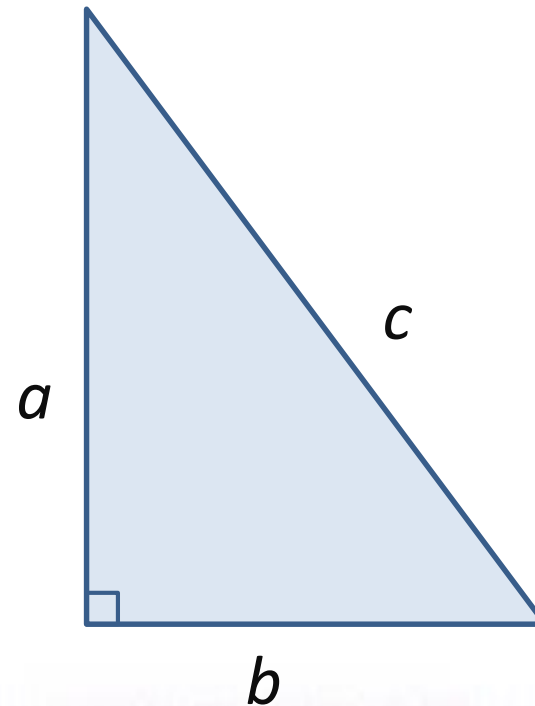
- Consistent (i.e., no statement in the system can be true and not true at the same time), and
- Capable of doing arithmetic of natural numbers (i.e., counting numbers),

there are always mathematical statements about natural numbers which are **true** but **unprovable** within the system

The Pythagorean Theorem

Theorem In any right-angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.

$$1. a^2 + b^2 = c^2$$

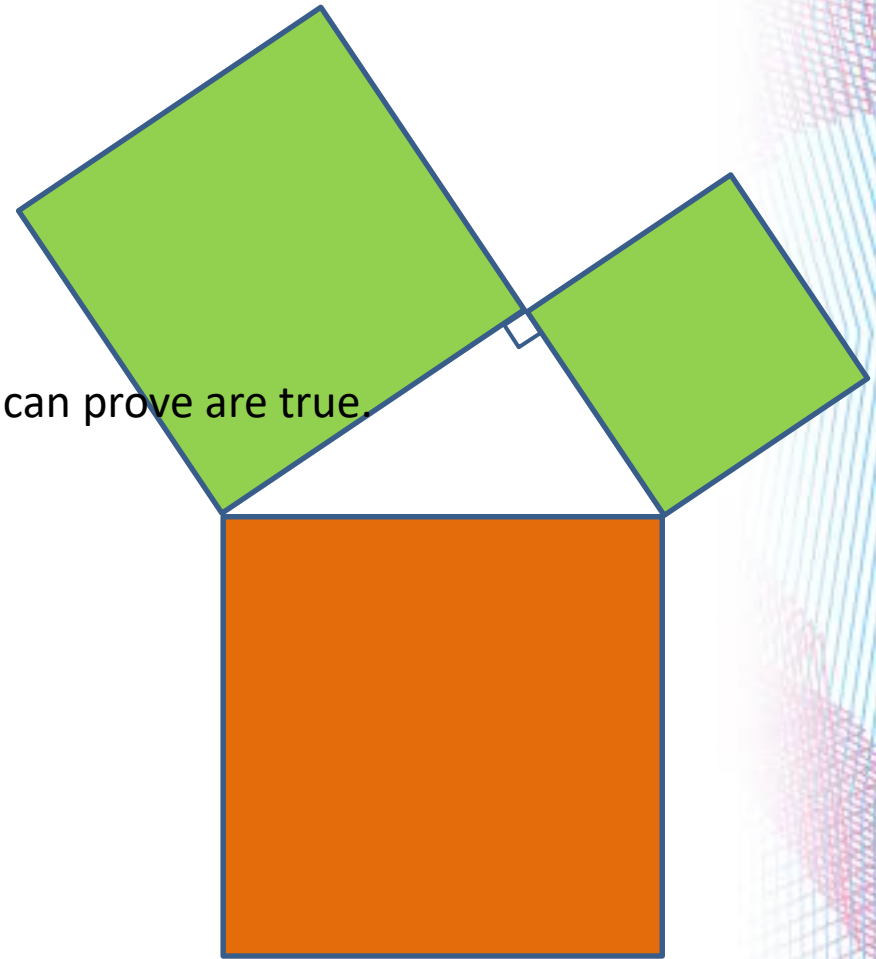


Euclid's *Elements*, Book I, Proposition 47

Proposition In right-angled triangles the square on the side subtending the right angle is equal to the squares on the sides containing the right angle.

Theorems are the things that we can prove are true.

(Here the squares are interpreted as **areas**)



Observing theorems

1. Looking for patterns is helpful for noticing what “appears” to be true.

$$2. 3^2 + 4^2 = 5^2, 5^2 + 12^2 = 13^2, 8^2 + 15^2 = 17^2, \dots$$

3. But does this mean that it is **always** true?

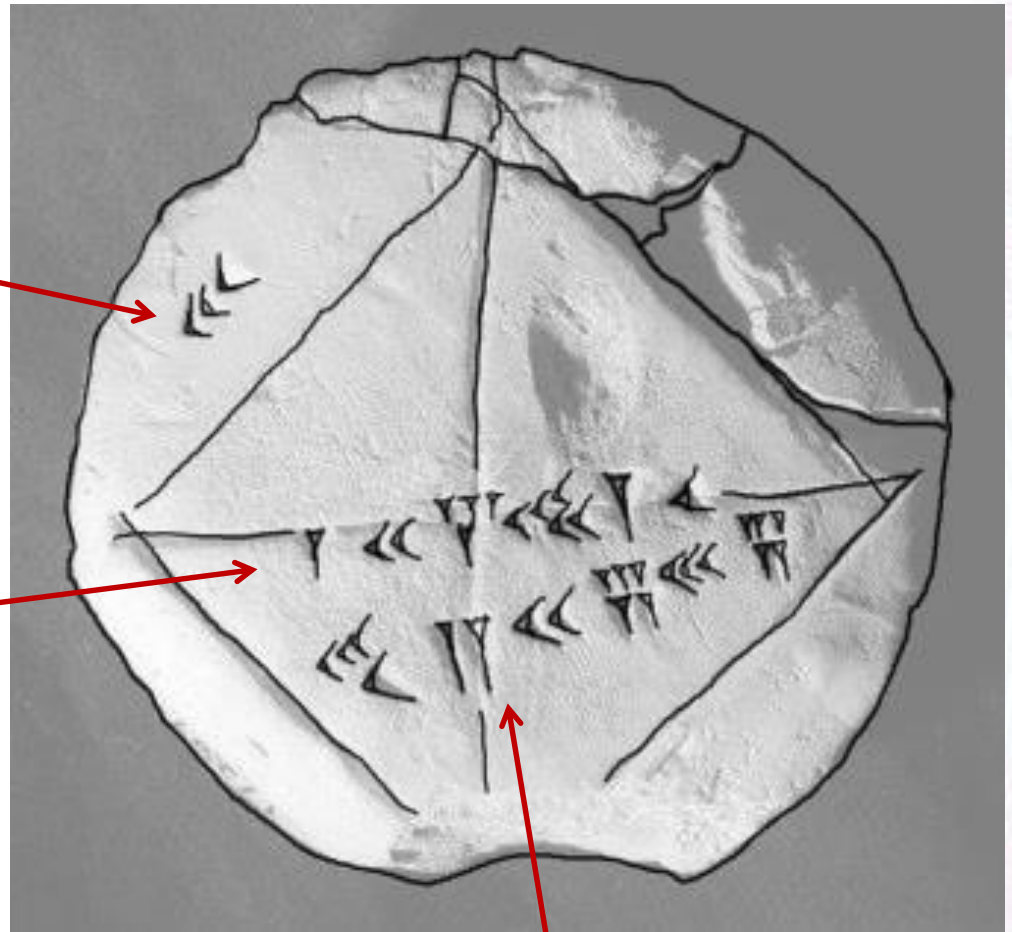
Long History of the Theorem

1. Babylonian tablet YBC 7289
2. Dated around 1,800-1,600 BC.

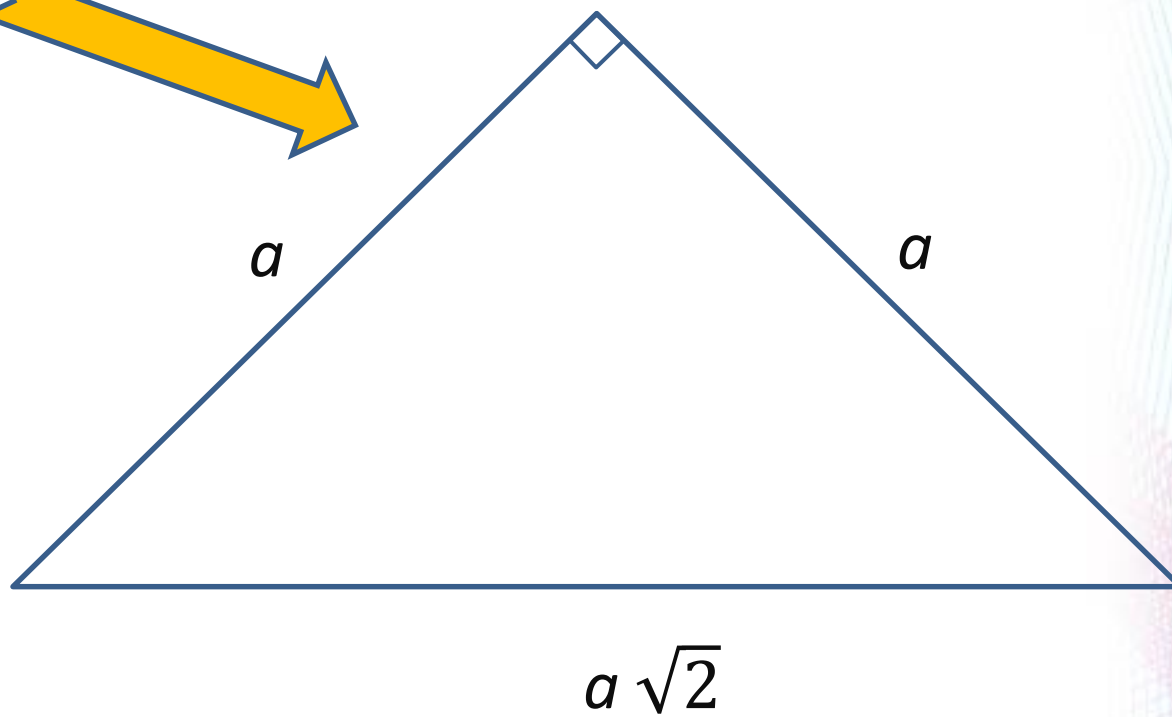
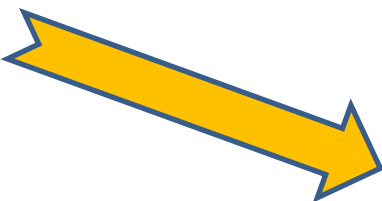
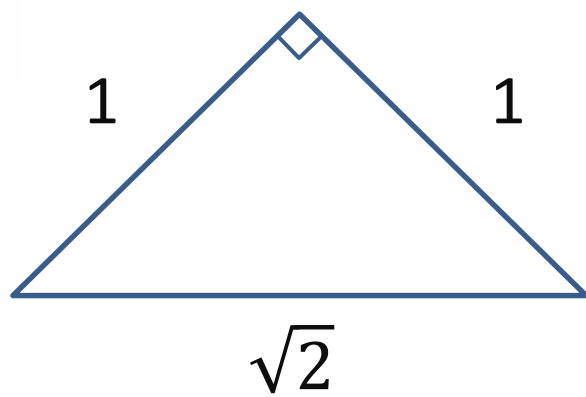


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$$\begin{aligned} &1,24,51,10 \\ &= 1 + 24/60 + 51/60^2 + 10/60^3 \\ &(\approx 1.414213) \\ &= \sqrt{2} \text{ (accurate to } 0.000,001) \end{aligned}$$



$$\begin{aligned} &42,25,35 \\ &= 42 + 25/60 + 35/60^2 \\ &(\approx 42.426389) \\ &= 30 \times \sqrt{2} \end{aligned}$$



Plimpton 322 Tablet

1. Babylonian tablet, dated around 1,800 BC.

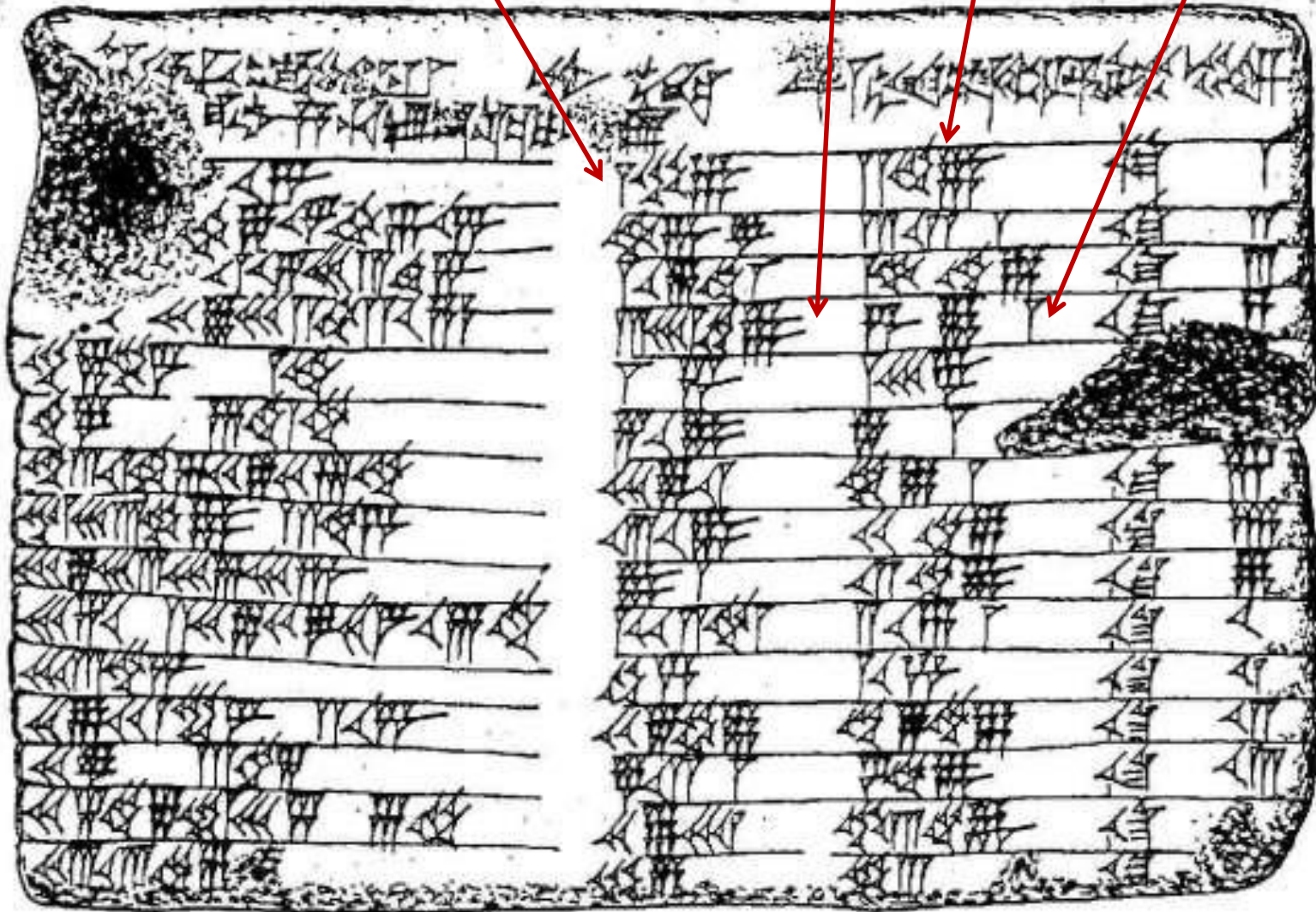


1,59

2,49

3,31,49

5,9,1



Implies $a = 13500$

12,709

18,541

Plimpton 322			
$(c/a)^2$	b	c	
[1,59,0,]15	1,59	2,49	1
[1,56,56,]58,14,50,6,15	56,7	3,12,1	2
[1,55,7,]41,15,33,45	1,16,41	1,50,49	3
[1,]5[3,1]0,29,32,52,16	3,31,49	5,9,1	4
[1,]48,54,1,40	1,5	1,37	5
[1,]47,6,41,40	5,19	8,1	6
[1,]43,11,56,28,26,40	38,11	59,1	7
[1,]41,33,59,3,45	13,19	20,49	8
[1,]38,33,36,36	9,1	12,49	9
1,35,10,2,28,27,24,26,40	1,22,41	2,16,1	10
1,33,45	45	1,15	11
1,29,21,54,2,15	27,59	48,49	12
[1,]27,0,3,45	7,12,1	4,49	13
1,25,48,51,35,6,40	29,31	53,49	14
[1,]23,13,46,40	56	53	15

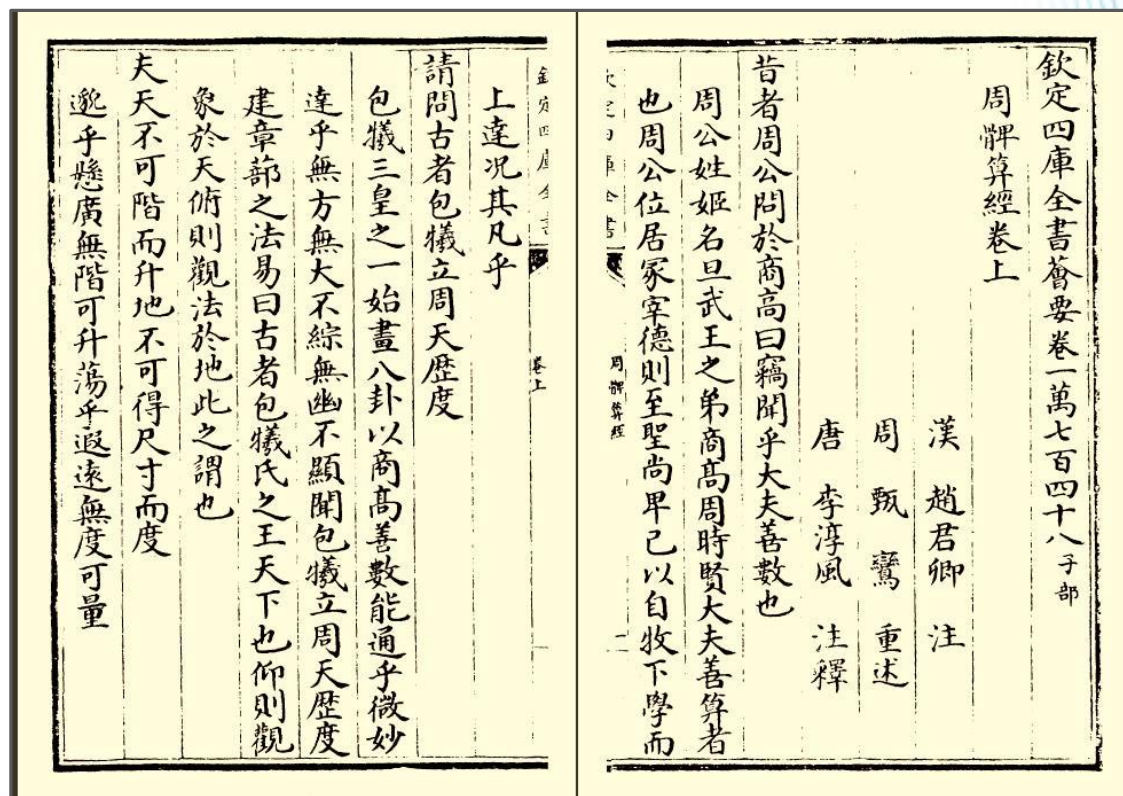
Note: The numbers in brackets are reconstructed.

Table relates to **Pythagorean triplets** : (a,b,c) where $a^2 + b^2 = c^2$

The (3,4,5) Triplet in Chinese Treatise

1. 周髀算經 *Zhou Bi Suan Jing* (“The Arithmetical Classic of the Gnomon and the Circular Paths of Heaven”)

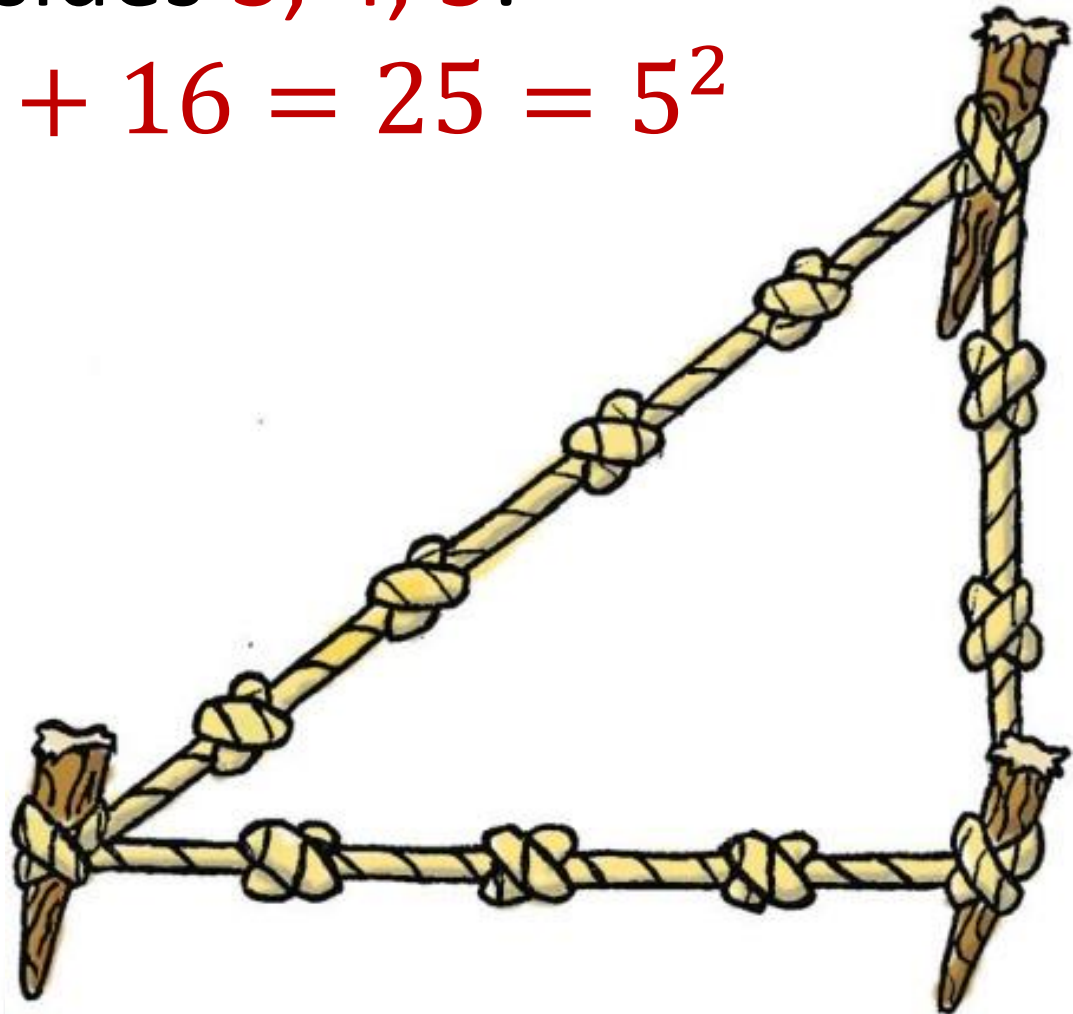
- Found during Western Han dynasty, but claimed as authored during the Zhou Dynasty
- Mentioned “勾三股四弦五” (right-angled triangle with sides 3, 4, and 5).



The Triplets (3, 4, 5)

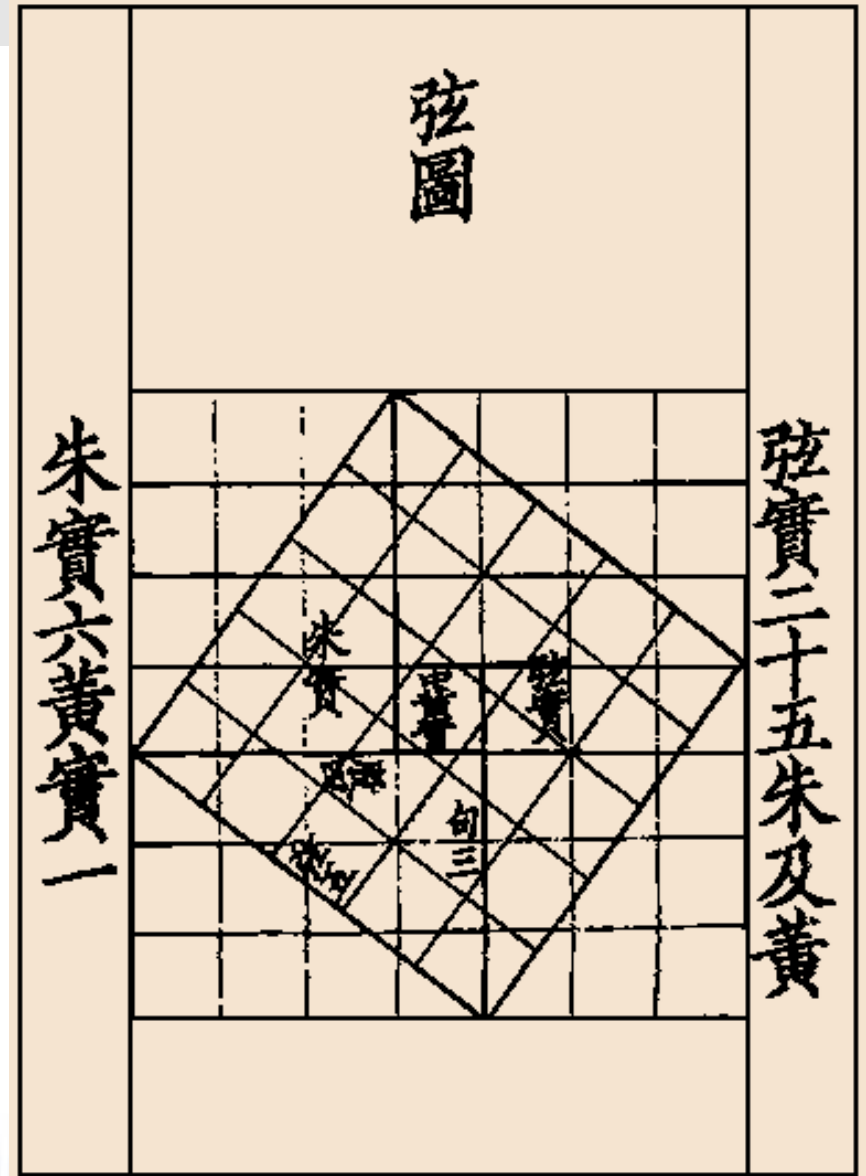
- Triangle with sides 3, 4, 5:

$$3^2 + 4^2 = 9 + 16 = 25 = 5^2$$



Gao Shuang Theorem

1. This diagram appeared in *Zhou Bi Suan Jing* gives a proof of the **Gao Shuang Theorem** (i.e., the Pythagorean Theorem).



Video: When not knowing Math can cost you \$15,000



Source: www.youtube.com/watch?v=BbX44YSsQ2I